Schneider Datathon Challenge

–Explainability on Classifier Models–

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01 MODEL PERFORMANCE SUMMARY

02 EXPLAINABILITY TECHNIQUE PROPOSAL AND INSIGHTS

01

MODEL PERFORMANCE SUMMARY

Random Forest

Logistic Regression

K-Nearest Neighbors

Random Forest Classifier

Support Vector Classifier SVC

Decision Tree Classifier

Gradient Boosting

Logistic Regression	0.58 – 0.6
K-Nearest Neighbors	0.75
Random Forest Classifier	0.84 – 0.86
Support Vector Classifier SVC	0.61
Decision Tree Classifier	0.81
Gradient Boosting	0.81 – 0.84

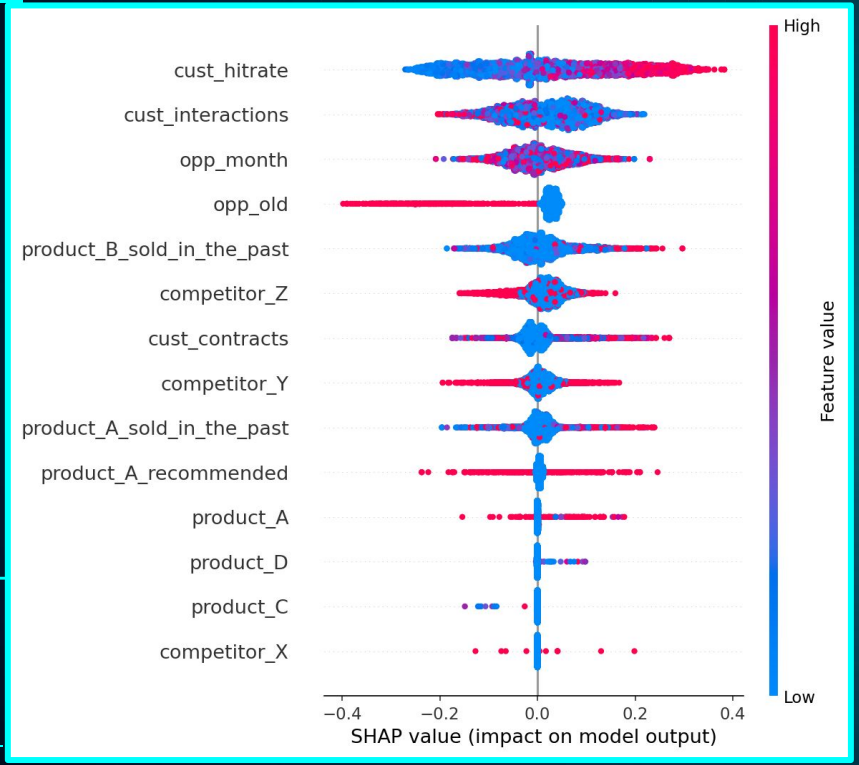
Logistic Regression	0.58 – 0.6
K-Nearest Neighbors	0.75
Random Forest Classifier	0.84 – 0.86
Support Vector Classifier SVC	0.61
Decision Tree Classifier	0.81
Gradient Boosting	0.81 – 0.84

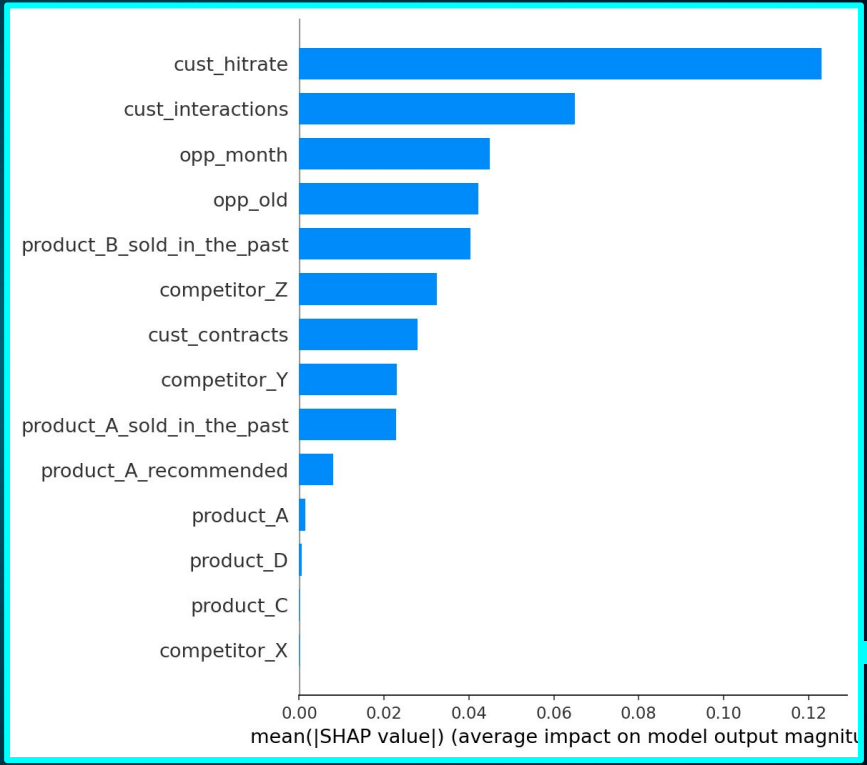
02

EXPLAINABILITY TECHNIQUE PROPOSAL AND INSIGHTS

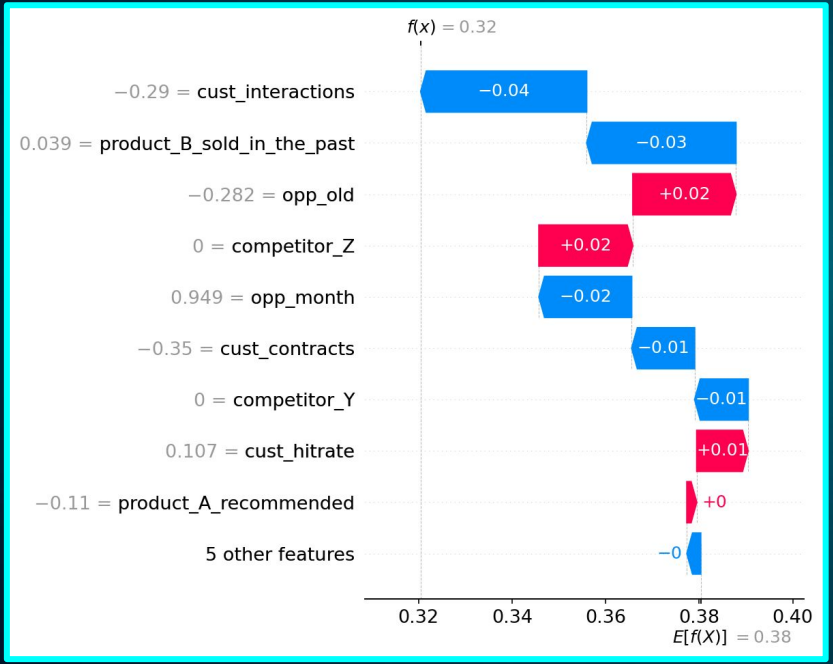
SHAP VALUES

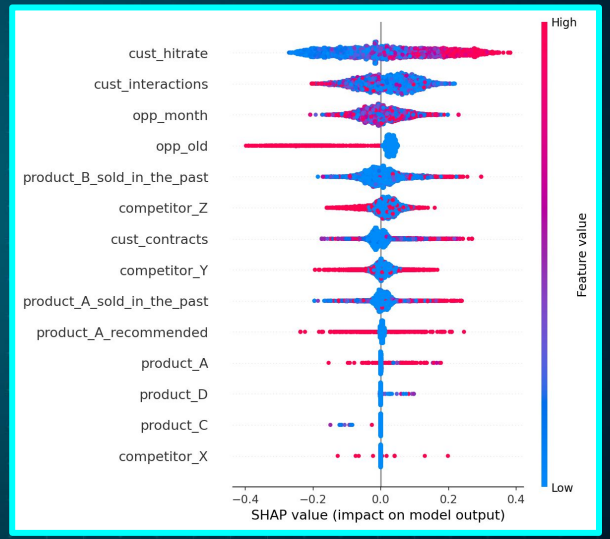
Global Graphics



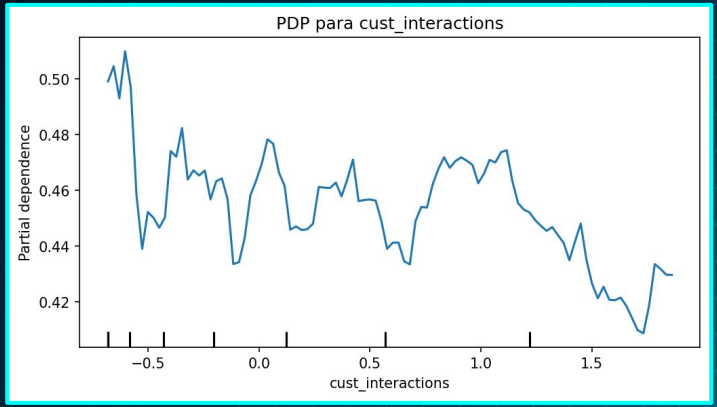


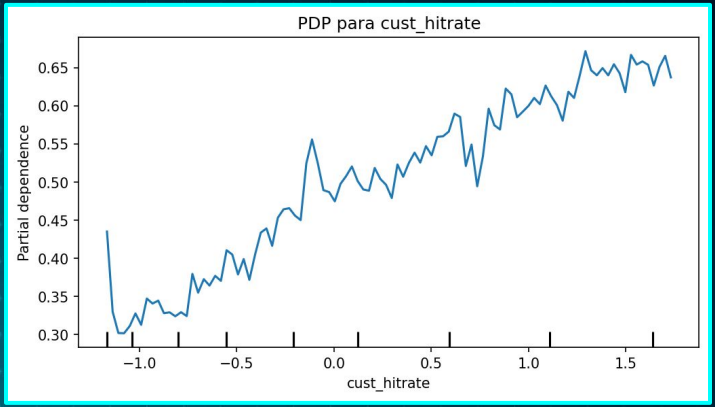
Local Graphic

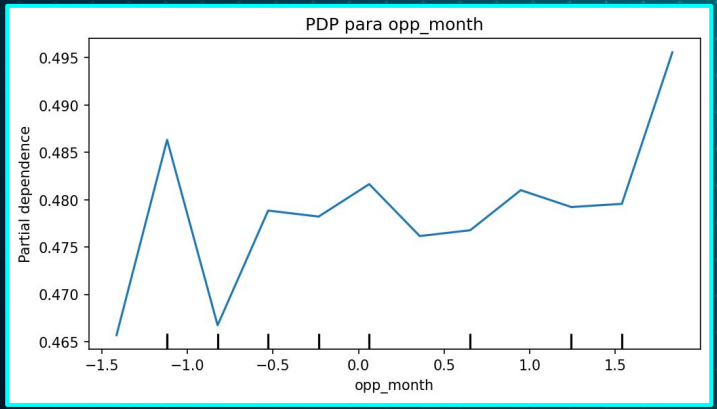


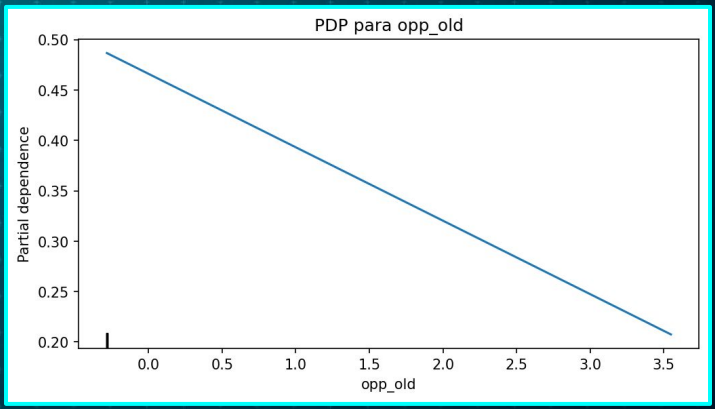


Partial Dependence Plots (PDP)


PDP para cust_interactions
Partial dependence
0.50
0.48
0.46
0.44
0.42
−0.5
0.0
0.5
1.0
1.5
cust_interactions


PDP para cust_hitrate
Partial dependence
0.65
0.60
0.55
0.50
0.45
0.40
0.35
0.30
−1.0
−0.5
0.0
0.5
1.0
1.5
cust_hitrate


PDP para opp_month
Partial dependence
0.495
0.490
0.485
0.480
0.475
0.470
0.465
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
opp_month


PDP para opp_old
Partial dependence
0.50
0.45
0.40
0.35
0.30
0.25
0.20
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
opp_old

THANKS FOR YOUR TIME